

1 Raw sources

Kaggle yield table
28,242 rows
Pakistan: 1,449 rows

**District soil and
climate dataset**
2,200 rows, 13 crops

2 Cleaning and join

Inner join with
district means by crop
wheat, maize:
414 rows

Group means by crop
(one value per crop)

3 Audit and collapse

**Audit: 9 copies of
each crop-year (one
yield each); collapse to
46 rows**

4 Modelling inputs

Feature sets
A: 9 tabular features
B: A + health score
C: A + year

Evaluation splits
random 5-fold
leave-one-year-out
forward chaining
(46- and 70-row tables;
season-wise for districts)

Replication data (Sections 5.5 and 5.6)

FAOSTAT, Pakistan
yield, pesticides,
temperature change

Wheat + maize,
1990-2024:
70 crop-year rows

District wheat yields
Punjab (PBS), 2007-20
37 rows, 13 seasons

Drop Islamabad, 2 missing
and 2 implausible cells:
464 observations

Leaf-image lane

PlantVillage
maize images: 3,852
3 disease classes ->
diseased vs healthy

80/20 split per class
(3,080 / 772); augment;
224x224; MobileNetV2
head training

Class-mean scores
 $s_h = 0.9921$, $s_d = 0.0007$
min-max rescale of
NPK_total -> health score

health score = feature B